

Student Number: _____ Class Teacher: _____

St George Girls High School

Trial Higher School Certificate Examination

2016



Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 15, show relevant mathematical reasoning and/or calculations

Section I	/10
Section II	
Question 11	/18
Question 12	/18
Question 13	/18
Question 14	/18
Question 15	/18
Total	/100

Total Marks – 100

Section I Pages 2 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided at the back of this paper

Section II Pages 7 – 16

90 marks

- Attempt Questions 11 – 15
- Allow about 2 hours and 45 minutes for this section
- Begin each question in a new writing booklet

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Section I

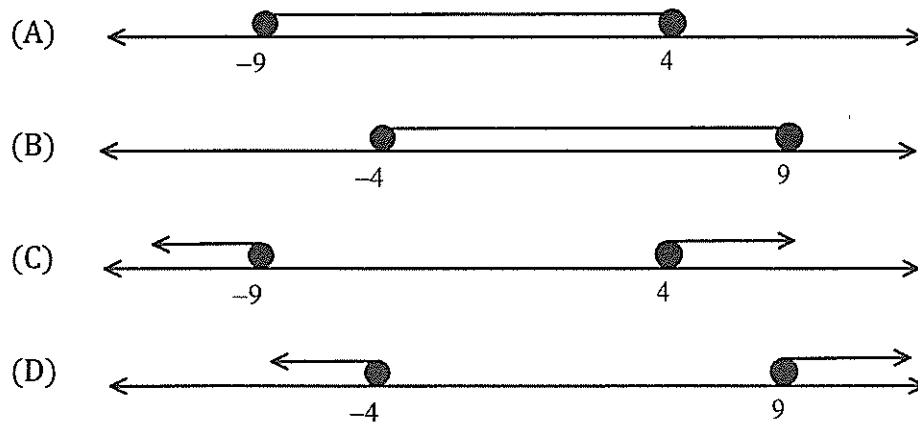
10 marks

Attempt Questions 1 - 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. Which graph shows the solution to $|2x - 5| \leq 13$?



2. What is the value of $\lim_{x \rightarrow 7} \frac{x^2 + 5x - 84}{x - 7}$?

- (A) -5
- (B) 0
- (C) 12
- (D) 19

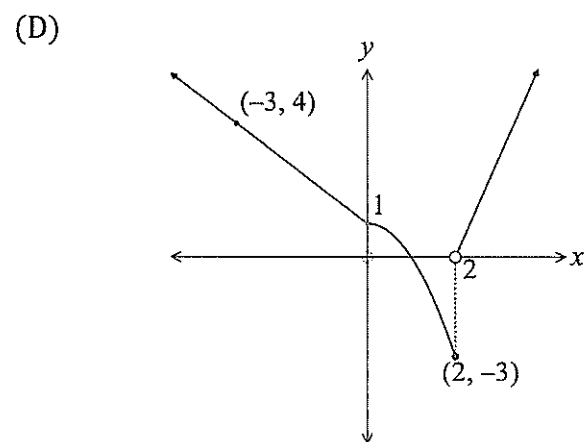
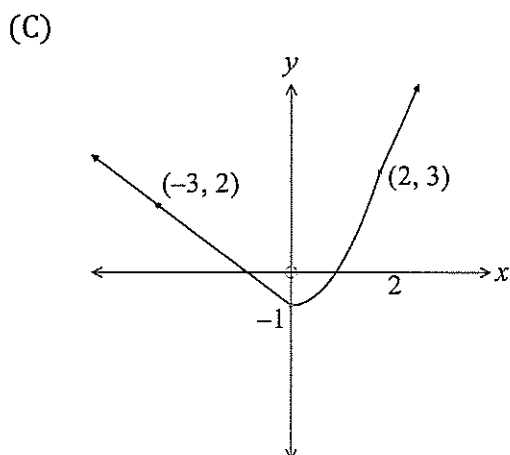
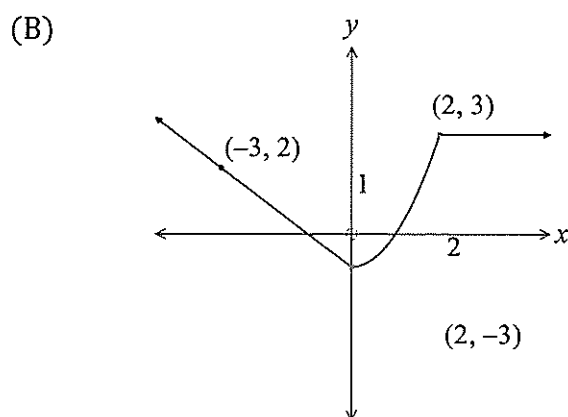
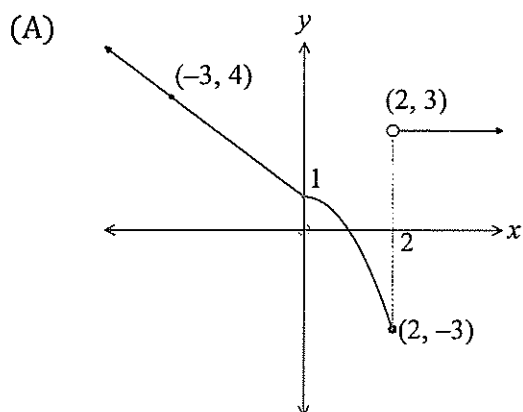
Section I (continued)

3. What is the value of $\sum_{r=1}^4 2^{1-r}$?
- (A) $\frac{1}{64}$
- (B) $\frac{9}{8}$
- (C) $\frac{15}{8}$
- (D) $\frac{9}{4}$
4. Which of the following is equivalent to $\operatorname{cosec}(\pi + \theta)$?
- (A) $\frac{-1}{\sin \theta}$
- (B) $\frac{-1}{\cos \theta}$
- (C) $\frac{1}{\cos \theta}$
- (D) $\frac{1}{\sin \theta}$

Section I (continued)

5. Which is the correct graph of the function below?

$$\begin{cases} y = 1 - x & x < 0 \\ y = 1 - x^2 & 0 \leq x \leq 2 \\ y = 3 & x > 2 \end{cases}$$



Section I (continued)

6. What is the solution to the equation $\log_e(x+2) - \log_e x = \log_e 4$?

- (A) $\frac{2}{5}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

7. The gradient function of a curve is $\frac{dy}{dx} = 3 - \frac{2}{x^2}$.

What is the equation of the curve if it passes through the point $(1, -2)$?

- (A) $y = \frac{4}{x^3}$
- (B) $y = \frac{2}{x} - 4$
- (C) $y = 3x - \frac{2}{x} - 3$
- (D) $y = 3x + \frac{2}{x} - 7$

8. What is the domain and range of the function $y = \frac{1}{\sqrt{x-9}}$?

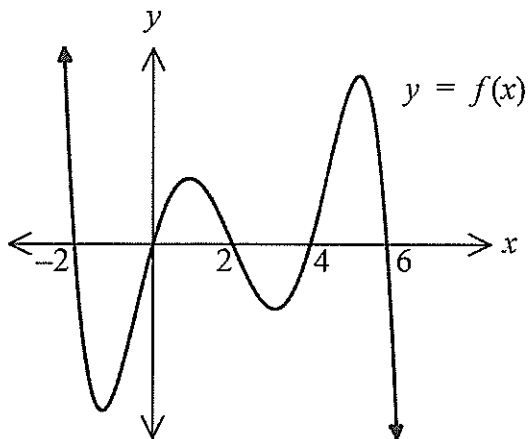
- (A) $\{x : x \geq 9\}$ and $\{y : y > 0\}$
- (B) $\{x : x > 9\}$ and $\{y : y > 0\}$
- (C) $\{x : -\infty \leq x \leq \infty\}$ and $\{y : -\infty \leq y \leq \infty\}$
- (D) $\{x : -3 \geq x \geq 3\}$ and $\{y : y < 0\}$

9. What are the amplitude and period of the function $f(x) = 2 - \sin 2x$?

- (A) Amplitude 1, period π
- (B) Amplitude 1, period 2π
- (C) Amplitude 2, period π
- (D) Amplitude 2, period 2π

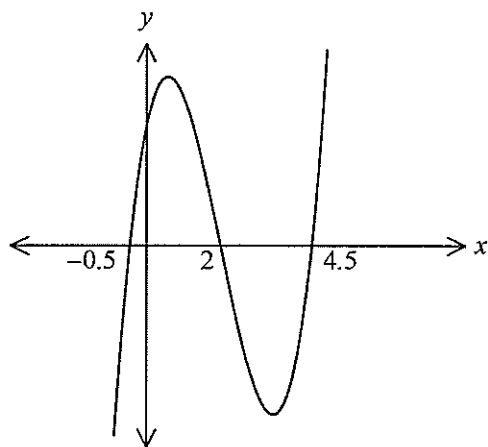
Section I (continued)

10. The graph of $y = f(x)$ is shown below.

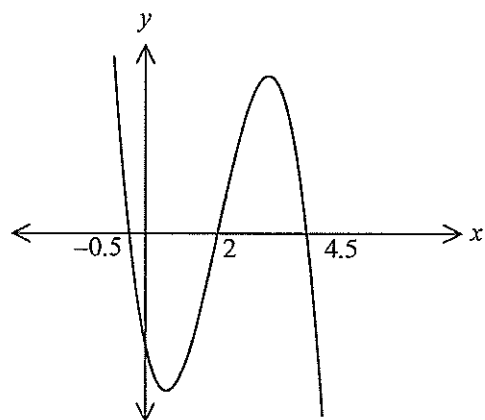


Which of these graphs represents $y = f'(x)$?

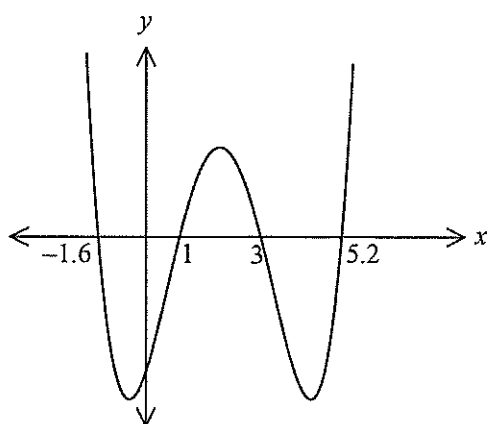
(A)



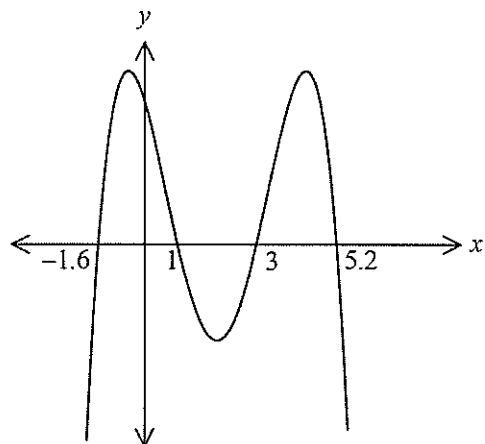
(B)



(C)



(D)



End of Section I

Section II

90 marks

Attempt Questions 11 – 15

Allow about 2 hours and 45 minutes for this section

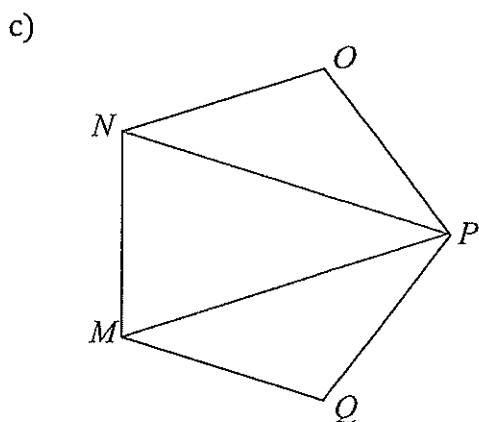
Start each question in a new writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11	(18 marks) Start a New Writing Booklet.	Marks
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a)	Simplify $\frac{5}{x-2} - \frac{2}{x-3}$.	2
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b)	Express $\frac{1}{\sqrt{5}-2}$ with a rational denominator.	2
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The diagram shows a regular pentagon $MNOPQ$.

- | | | |
|-------|---|---|
| (i) | Show that triangle NOP is isosceles and hence find $\angle ONP$. | 3 |
| (ii) | Show that triangles NOP and PQM are congruent. | 2 |
| (iii) | Find the size of $\angle MPN$. | 1 |

Question 11 (continued)

Marks

- d) Boxes are stacked in layers, where each layer contains one box less than the layer below. There are six boxes in the top layer, seven boxes in the next layer, and so on. There are n layers altogether.

(i) Write down the number of boxes in the bottom layer.

2

(ii) Show that there are $\frac{1}{2}n(n+11)$ boxes.

2

- e) A particle moves along the x -axis with acceleration $3t-2$. Initially it is 4 units to the right of the origin, with a velocity of 2 units per second.
What is the position of the particle after 5 seconds?

3

- f) Find $\int \frac{x}{x^2+3} dx$.

1

Question 12 (18 marks) Start a New Writing Booklet.

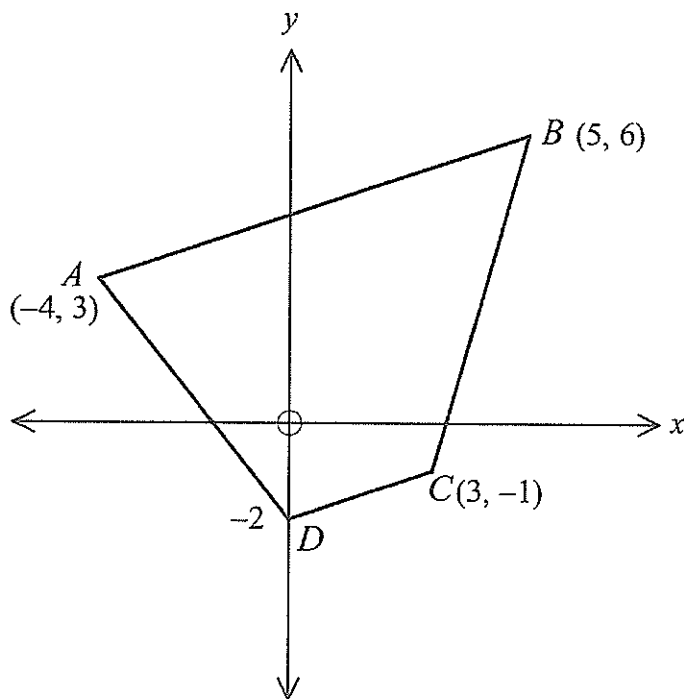
Marks

- a) Consider the series $S_n = -18 + (-15) + (-12) + \dots + u_n$

2

For what value of n is $S_n = 0$?

- b) A quadrilateral is formed by the points $A(-4, 3)$, $B(5, 6)$, $C(3, -1)$ and $D(0, -2)$ as shown in the diagram.



- (i) Show that the quadrilateral is a trapezium, with $AB \parallel DC$.

2

- (ii) Show that the equation of AB is $x - 3y + 13 = 0$.

1

- (iii) Find the perpendicular distance from D to AB .

1

- (iv) Find the area of the trapezium $ABCD$.

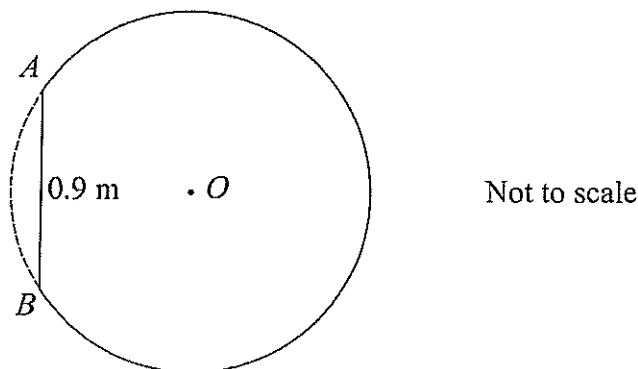
2

Question 12 (continued)	Marks
c) Evaluate $\int_0^{\frac{\pi}{3}} \cos 2x dx$.	2
d) Consider the curve $y = 2x^3 + 3x^2 - 36x + 4$ for $-5 \leq x \leq 5$.	
(i) Find the stationary points and determine their nature.	3
(ii) Find the point of inflexion.	2
(iii) Sketch the curve for $-5 \leq x \leq 5$.	2
(iv) Find the maximum value in the domain given.	1

Question 13 (18 marks) Start a New Writing Booklet.

Marks

- a) The circle has a centre O and radius 0.9 metres.
A small segment of this circle has been removed as shown below.
The length of the straight edge AB is also 0.9 metres.



- (i) Explain why $\angle AOB = \frac{\pi}{3}$. 1
- (ii) What is the shaded area to 3 significant figures? 3
- b) Find the value of k if the sum of the roots of $x^2 - (k-1)x + 2k = 0$ is equal to the product of the roots. 2
- c) Prove that $\frac{1}{\sec A - 1} + \frac{1}{\sec A + 1} = 2 \cot A \operatorname{cosec} A$. 3

Question 13 (continued)

Marks

- d) The second term of an arithmetic series is 37 and the sixth term is 17.
What is the sum of the first ten terms? 3
- e) Find the volume when $y = \log_e x$ is rotated about the y -axis between $y = 1$
and $y = 3$. Express your answer in exact form. 3
- f) Differentiate with respect to x .
- (i) $(e^x - 3)^4$ 1
- (ii) $x \tan x$ 1
- (iii) $\log_e(\cos x)$ 1

Question 14 (18 marks) Start a New Writing Booklet.

Marks

- a) Find the equation of the normal to the curve $y = x \log_e x$ at the point on the curve where $x = 1$.

3

- b) Twenty kilograms of sugar is placed in a container of water and begins to dissolve. After t hours the amount A kg of undissolved sugar is $A = 20e^{-kt}$.

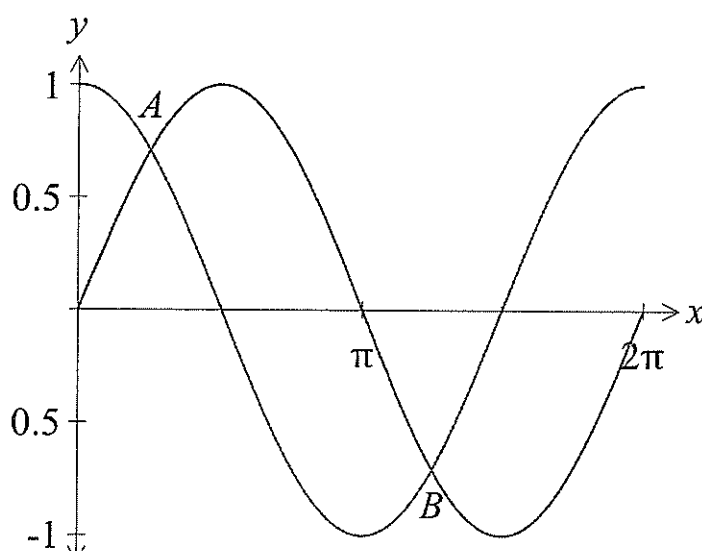
- (i) Calculate k , given that $A = 4.8$ when $t = 5$.

1

- (ii) After how many hours does 1 kg of sugar remain undissolved?

2

c)



The diagram above shows the graphs $y = \sin x$ and $y = \cos x$ in the domain $0 \leq x \leq 2\pi$. The graphs intersect at points A and B .

- (i) What are the coordinates of A and B ?

2

- (ii) Find the area enclosed by the two graphs between A and B .

2

Question 14 (continued)

Marks

- d) On 1 June 2006, Patrick invested \$20,000 in a bank account that paid interest at a fixed rate of 7% per annum, compounded annually.
- (i) How much would be in the account after the payment of interest on 1 June 2016 if no additional deposits were made? 1
- (ii) Patrick decided to add \$2000 to his account on 1 June each year beginning on 1 June 2007. How much is in his account after Patrick makes a final deposit on 1 June 2016? 3
- (iii) Patrick's friend Bella invested \$20,000 in an account at another bank on 1 June 2006 and made no further deposits. On 1 June 2016, the balance of Bella's account was \$49,565. What was the annual rate of compound interest paid on Bella's account? 2
- e) Differentiate $\log_2 x^2$. 2

Question 15 (18 marks) Start a New Writing Booklet.

Marks

- a) A spokesman made the following statement about immigration:
 “Over that period of time the number of immigrants was increasing at a decreasing rate”.

- (i) Given I is the level of immigration and t is time, what does the statement mean about

$$\frac{dI}{dt} \text{ and } \frac{d^2 I}{dt^2} ? \quad 1$$

- (ii) Sketch a graph of immigration against time that fits the above information. 1

- b) (i) Shade the intersection of these regions. 2

$$\begin{aligned} x &\leq 2 \\ y &\leq 3 \\ y &\geq |x| \end{aligned}$$

- (ii) Hence, find the area of the region defining the intersection. 1

- c) The speed of a train was recorded at intervals of one minute. The times, in minutes, and the corresponding speeds v , in kilometres per hour, are listed in the table below.

Time (min)	0	1	2	3	4
Speed (km/h)	0	24	35	28	50

Given that the distance x , in km, travelled by the train, may be found by calculating the area under a velocity/time graph, use Simpson’s Rule with five function values to estimate the distance travelled by the train during the first four minutes.

3

Question 15 (continued)

Marks

d) A parabola has the equation $16y = x^2 - 4x - 12$.

(i) Find the coordinates of the vertex. 2

(ii) Find the coordinates of the focus. 1

(iii) Find the equation of the directrix. 1

e) A can is in the shape of a closed cylinder with a height h cm and a radius r cm.
The volume of the can is 200 cm^3 .

(i) Find an expression for h in terms of r . 1

(ii) Show that the surface area $S \text{ cm}^2$ of the can is given by the formula 1

$$S = 2\pi r^2 + \frac{400}{r} .$$

(iii) If the area of the metal used to make the can is to be minimized, find the area of the can to the nearest cm^2 . 4

End of Examination

MATHEMATICS TRIAL HSC 2016 – SECTION I – MULTIPLE CHOICE

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
① $2x - 5 \leq 13$ $\swarrow$ $\searrow$ $+(2x - 5) \leq 13$ $2x \leq 18$ $x \leq 9$ $-(2x - 5) \leq 13$ $2x - 5 \geq -13$ $2x \geq -8$ $x \geq -4$  $\therefore \textcircled{B}$		
② $\lim_{x \rightarrow 7} \frac{x^2 + 5x - 84}{x - 7}$ $= \lim_{x \rightarrow 7} \frac{(x + 12)(x - 7)}{x - 7}$ $= 7 + 12$ $= 19 \quad \therefore \textcircled{D}$		

MATHEMATICS – SECTION I – MULTIPLE CHOICE

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

③ $\sum_{r=1}^4 2^{1-r}$

$$= 2^0 + 2^{-1} + 2^{-2} + 2^{-3}$$

$$= 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$$

$$= 1\frac{7}{8}$$

$$= \frac{15}{8} \quad \therefore \textcircled{C}$$

④ $\operatorname{cosec}(\pi + \theta)$

$$= \frac{1}{\sin(\pi + \theta)}$$

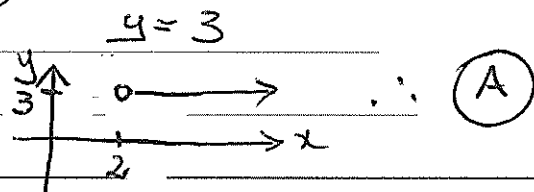
$$= \frac{1}{-\sin \theta}$$

$$= \frac{-1}{\sin \theta} \quad \therefore \textcircled{A}$$

Diagram: A circle with angles $180-\theta$, θ , $360-\theta$, and $180+\theta$. The quadrants are labeled S, A, C, T.

⑤ $y = -x + 1$ ~~ⓑ~~ ~~Ⓒ~~

Ⓐ OR Ⓓ



MATHEMATICS – SECTION I – MULTIPLE CHOICE

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$(6) \log_e(x+2) - \log_e x = \log_e 4$$

$$\frac{x+2}{x} = 4$$

$$x+2 = 4x$$

$$3x = 2$$

$$\therefore x = \frac{2}{3}$$

$\therefore (B)$

$$(7) \frac{dy}{dx} = 3 - 2x^{-2}$$

$$y = 3x - \frac{2x^{-1}}{-1} + C$$

$$y = 3x + \frac{2}{x} + C$$

subst. $(1, -2)$ to find C

$$-2 = 3 + 2 + C$$

$$-2 = 5 + C$$

$$C = -7$$

$$\therefore y = 3x + \frac{2}{x} - 7 \quad \therefore (D)$$

MATHEMATICS – SECTION I – MULTIPLE CHOICE

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

⑧ $y = \frac{1}{\sqrt{x-9}}$

$\sqrt{x-9} \neq 0$

$x-9 > 0$

$\therefore \Delta : x > 9$

$R : y > 0$

$\therefore \textcircled{B}$

⑨ $f(x) = 2 - \sin 2x$

$= -\sin 2x + 2$

amplitude = 1

period = $\frac{2\pi}{2} = \pi$

$\therefore \textcircled{A}$

⑩ turning points become x-intercepts

ie 4 x-intercepts

~~Ⓐ~~

~~Ⓑ~~

Ⓒ OR Ⓓ

between -1.6 & 1, $f'(x) > 0$

$\therefore \textcircled{D}$

10

MATHEMATICS - QUESTION NO: 11 continued

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) $\frac{5}{x-2} - \frac{2}{x-3}$ $= \frac{5(x-3) - 2(x-2)}{(x-2)(x-3)}$ $= \frac{5x-15-2x+4}{(x-2)(x-3)}$ (1 mark) $= \frac{3x-11}{(x-2)(x-3)}$ (1 mark)	1	generally very well done
b) $\frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} = \frac{\sqrt{5}+2}{(\sqrt{5})^2-4}$ (1 mark) $= \frac{\sqrt{5}+2}{1}$ $= \sqrt{5}+2$ (1 mark)	1	some mistakes were made by multiplying by $\sqrt{5}-2$ instead of $\sqrt{5}+2$
c) i) In the regular pentagon, $\triangle NOP$ is isosceles $[ON=OP]$ (1 mark) Interior angles of regular pentagon $= \frac{180 \times 5}{5}$ $= 108^\circ$ (1 mark) $\angle ONP = \frac{180^\circ - 108^\circ}{2}$ $= 36^\circ$ (1 mark)	1	generally well done
	1	

MATHEMATICS - QUESTION NO: 11 continued

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
ii) In $\triangle NOP$ and $\triangle PQN$ $\left. \begin{array}{l} ON = MQ \text{ [sides of regular pentagon]} \\ \angle NOP = \angle PQM \text{ [equal interior angles]} \\ OP = QP \text{ [sides of regular pentagon]} \end{array} \right\} \text{1 mark}$ $\therefore \triangle NOP \equiv \triangle PQN$ (SAS TEST) 1 mark	1	Students should tidy up their proofs and be more concise.
iii) $\angle OPQ = \angle OPN + \angle NPQ + \angle MPQ$ $108^\circ = 36^\circ + \angle NPQ + 36^\circ$ $\angle NPQ = 108^\circ - 72^\circ$ $= 36^\circ \quad (1 \text{ mark})$	1	well done.
d) i) $T_n = a + (n-1)d$ where $a=6, d=1$ — (1 mark) $= 6 + (n-1) \cdot 1$ $= n+5 \quad (1 \text{ mark}) \quad \text{bottom layer.}$	1	well done.
ii) Total number of boxes $S_n = \frac{n}{2} \{2a + (n-1)d\} \quad \text{or} \quad \frac{n}{2} \{a+l\}$ $= \frac{n}{2} \{12 + (n-1)1\} \quad = \frac{n}{2} \{6 + n+5\} \quad (1 \text{ mark})$ $= \frac{n}{2} \{n+11\} \quad = \frac{n}{2} \{n+11\} \quad (1 \text{ mark})$	1	well done

MATHEMATICS - QUESTION NO: 11 continued

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
e) $a = 3t - 2$ $\dot{x} = 3t^2 - 2t + c$ when $t=0$, $\dot{x} = 2 \therefore c = 2$ $\dot{x} = \frac{3t^2}{2} - 2t + 2$ (1 mark)	1	ONLY MISTAKE MADE IN THIS PART WAS NEGLECTING THE CONSTANT.
$x = \frac{t^3}{2} - t^2 + 2t + c$ when $t=0$, $x = 4 \therefore c = 4$		SINCE IT WAS CRUCIAL MARKS WERE LOST EACH
$x = \frac{t^3}{2} - t^2 + 2t + 4$ (1 mark)	1	TIME!
when $t=5$		
$x = \frac{5^3}{2} - 5^2 + 10 + 4$		
$= 51.5$		
particle is 51.5 units to the right after 5 seconds. (1 mark)	1	MANY students left off the constant.
f) $\frac{1}{2} \int \frac{2x}{x^2+3} dx = \frac{1}{2} \ln(x^2+3) + c$		but no marks were taken off if they had included constant in part (e)

MATHEMATICS - QUESTION NO: 12

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

i) (a) $S_n = (-18) + (-15) + (-12) + \dots + u_n$

$$a = -18 \quad d = 3$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$0 = \frac{n}{2} [2 \times -18 + (n-1) \times 3]$$

$$0 = \frac{n}{2} (-36 + 3n - 3)$$

$$0 = n(3n - 39)$$

$$n = 0 \quad \text{or} \quad n = 13$$

But $n \neq 0$ as $n > 0$, thus $n = 13$

1

I only gave them $1\frac{1}{2}$ if they did not mention $n=0$ as a soln to $n(3n-39)=0$

(b) $m_{DC} = \frac{-1-2}{3-0} = \frac{1}{3} = m_1$

$$m_{AB} = \frac{6-3}{5-4} = \frac{1}{3} = m_2$$

$\therefore AB \parallel DC$ since $m_1 = m_2$

$\therefore ABCD$ is a trapezium (at least one pair sides parallel)

1

NOTE: A lot of students, nearly all thought a trapezium had only one pair parallel sides

$\frac{1}{2}$

I took $\frac{1}{2}$ off for not writing the conclusion, first part, (last line)

$\frac{1}{2}$

ii)

$$m = \frac{1}{3}$$

Either $y - 6 = \frac{1}{3}(x - 5)$

$$3y - 18 = x - 5$$

$$x - 3y + 13 = 0$$

(OR) $y - 3 = \frac{1}{3}(x + 4)$

$$3y - 9 = x + 4$$

$$x - 3y + 13 = 0$$

1

Most students got the mark for this

MATHEMATICS - QUESTION NO: 12 continued

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(iii) $d = \left \frac{ax_1 + by_1 + c_1}{\sqrt{a^2 + b^2}} \right$ ℓ $x - 3y + 13 = 0$ $D(0, -2)$ $= \left \frac{1 \times 0 - 3 \times -2 + 13}{\sqrt{1^2 + (-3)^2}} \right$ $= \frac{19}{\sqrt{10}}$	1	Some did $1 \times 0 = 1$ and so got $\frac{20}{\sqrt{10}} = 2\sqrt{10}$ penalised $\frac{1}{2}$ a mark.
(iv) $AB = \sqrt{90}$ $DC = \sqrt{10}$ $= 3\sqrt{10}$ $A = \frac{1}{2}h(a+b)$ $= \frac{1}{2} \times \frac{19}{\sqrt{10}} (3\sqrt{10} + \sqrt{10})$ $= 38 \text{ u}^2$	1	$\frac{1}{2}$ mark each
(c) $\int_0^{\pi/3} \cos 2x \, dx$ $= \frac{1}{2} [\sin 2x]_0^{\pi/3}$ $= \frac{1}{2} [\sin \frac{2\pi}{3} - \sin 0]$ $= \frac{1}{2} (\frac{\sqrt{3}}{2} - 0)$ $= \frac{\sqrt{3}}{4}$	1	Full marks given if incorrect answer to (iii) was carried forward correctly.
	1	Most did this correctly, some did not write $\sin 0$ but I did not penalise them

MATHEMATICS - QUESTION NO: 12 continued

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

(d) (i) $\frac{dy}{dx} = 6x^2 + 6x - 36$

Stationary points occur when $\frac{dy}{dx} = 0$

$$6(x+3)(x-2) = 0$$

$$x = -3 \quad x = 2$$

$$y = 85 \quad y = -40$$

$$\frac{d^2y}{dx^2} = 12x + 6$$

When $x = -3$ $\frac{d^2y}{dx^2} = -30 < 0$ thus $(-3, 85)$ is a max. turning point

When $x = 2$ $\frac{d^2y}{dx^2} = 30 > 0$ thus $(2, -40)$ is a min. turning point

(OR) could do table of values for x and y'

(ii) Possible point of inflexion when $\frac{d^2y}{dx^2} = 0$

$$12x + 6 = 0$$

$$x = -\frac{1}{2}, y = 22\frac{1}{2}$$

x	-1	$-\frac{1}{2}$	0
$\frac{d^2y}{dx^2}$	-6	0	6

Since change in concavity

$(-\frac{1}{2}, 22\frac{1}{2})$ is a point of inflexion

1

1

$\frac{1}{2}$ for each correct set of values.

1

Some did not put values in for $\frac{d^2y}{dx^2}$, not many, and were penalised a mark.

1

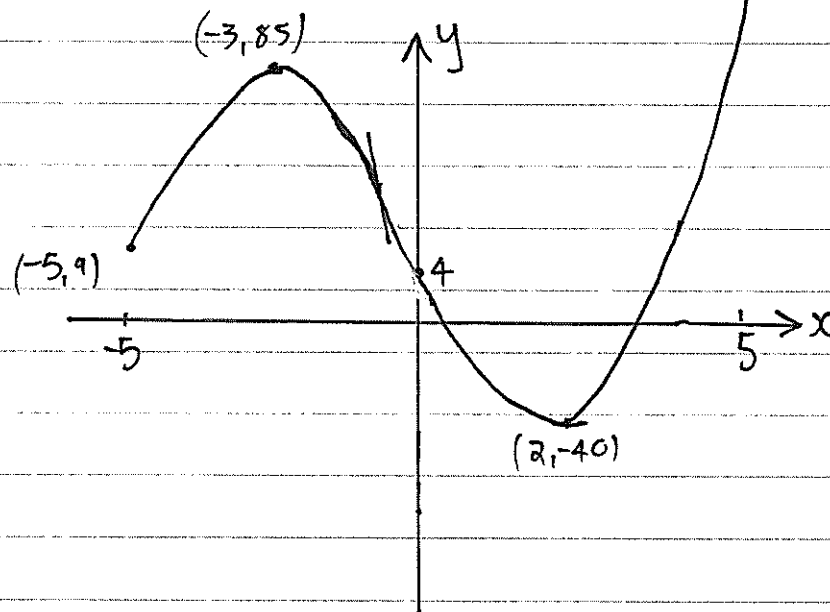
MANY did not show concavity change in the table and just assumed it was a point of inflexion

1

MATHEMATICS - QUESTION NO: 12 continued

SUGGESTED SOLUTIONS

(iii.)



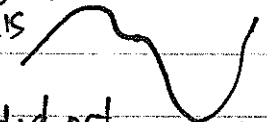
$\frac{1}{2}$ shape +
 $\frac{1}{2}$ y intercept
 $\frac{1}{2}$ end pts
 $\frac{1}{2}$ turning points
 $\frac{1}{2}$ inflexion point

MARKS

MARKER'S COMMENTS

2

Many thought a point of inflexion had to be horizontal and thus their graph looked like this



Some did not show y intercept and I penalised them $\frac{1}{2}$ (this was what was in the original guidelines)

(iv.) Maximum value is 149 when $x = 5$

$\frac{1}{2}$ given for (5, 149)
 $\frac{1}{2}$ given for $x = 149$ (generous).

MATHEMATICS - QUESTION NO: 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
Q 13 (a) (i) $OA = OB = AB = 0.9$ $\therefore \triangle AOB$ is equilateral $\therefore \angle AOB = \frac{180^\circ}{3} = 60^\circ = \frac{\pi}{3}$	1	
(ii) Shaded area = $\pi \times 0.9^2 - \left[\frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta \right]$ $= 0.81\pi - \left[\frac{1}{2} \times 0.9^2 \times \frac{\pi}{3} - \frac{1}{2} \times 0.9^2 \times \sin \frac{\pi}{3} \right]$ $\xrightarrow{\text{①}} \quad \quad \quad \xrightarrow{\text{①}}$ $= 0.81\pi - \left[0.135\pi - 0.405 \times \frac{\sqrt{3}}{2} \right]$ $\doteq 2.4713$ $\doteq 2.47 \text{ m}^2$	1 1	$-\frac{1}{2}$ mark of any minor error.
(b) $a + b = -\frac{b}{a}$ $a + b = \frac{c}{a}$ $= -\frac{(k-1)}{1}$ $= \frac{2k}{1}$ $\therefore (k-1) = 2k$ $k = -1$	1 1	-1 for $a + b = -\frac{k+1}{1-k}$ many identified $b = (b-1)$

MATHEMATICS - QUESTION NO: 13 continued

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
Q13 (c) $\frac{1}{\sec A - 1} + \frac{1}{\sec A + 1} = 2 \cot A \operatorname{cosec} A$		
$\text{LHS} = \frac{\sec A + 1 + \sec A - 1}{\sec^2 A - 1}$	1	-1/2 mark for any minor error.
$= \frac{2 \sec A}{\tan^2 A}$		
$= 2 \times \frac{1}{\cancel{\cos A}} \times \frac{\cos^2 A}{\sec^2 A}$	1	
$= 2 \times \frac{\cos A}{\sin A} \times \frac{1}{\sin A}$	1	
$= 2 \cot A \operatorname{cosec} A$		
$= \text{RHS.}$		
(d) $T_2 = a + d = 37$ —①		
$T_6 = a + 5d = 17$ —②		-1/2 mark for minor error.
②-① $4d = -20$		
$d = -5$	1	
substituting $d = -5$ into ①		
$a - 5 = 37$		
$a = 42$	1	
$S_{10} = \frac{10}{2} [2(42) + 9 \times (-5)]$		
$= 195$	1	

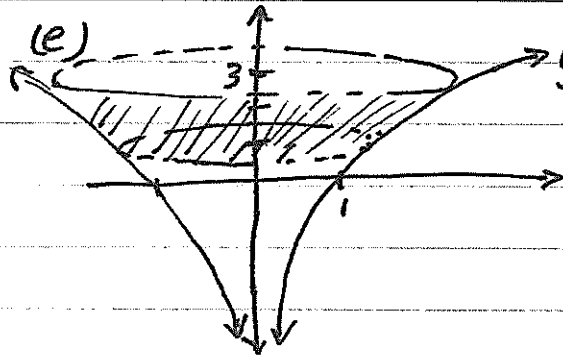
MATHEMATICS - QUESTION NO: 13 continued

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

Q13 (e)



$$y = \log_e x$$

$$y = \log_e x$$

$$x = e^y$$

$$V = \pi \int_a^b x^2 dy.$$

$$= \pi \int_1^3 (e^y)^2 dy.$$

$$= \frac{\pi}{2} [e^{2y}]_1^3$$

$$= \frac{\pi}{2} [e^6 - e^2]$$

1

-1/2 mark for any minor error.

1

1

(f) (i) $y = (e^x - 3)^4$

$$\frac{dy}{dx} = 4(e^x - 3)^3 \times e^x$$

$$= 4e^x (e^x - 3)^3$$

1

(ii) $y = x \tan x$

$$\frac{dy}{dx} = x \cdot \sec^2 x + \tan x \cdot 1$$

$$= x \sec^2 x + \tan x$$

1

(iii) $y = \log_e (\cos x)$

$$\frac{dy}{dx} = \frac{1}{\cos x} \times -\sin x$$

$$= -\frac{\sin x}{\cos x}$$

$$= -\tan x$$

1

MATHEMATICS - QUESTION NO: 14

	SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a)	$y = x \log_e x$ $u = x$ $u' = 1$ $v = \log_e x$ $v' = \frac{1}{x}$ $y' = x \times \frac{1}{x} + 1 \times \log_e x$ $= 1 + \log_e x$ When $x=1$, $y=0$, $y'=1$ $\therefore$ gradient of normal $= -1$ $y - 0 = -1(x - 1)$ $y = -x + 1$	1 1 1	Evaluate the gradient of the tangent before finding the negative reciprocal.
b) i	$4.8 = 20e^{-5k}$ $e^{-5k} = 0.24$ $-5k = \ln 0.24$ $k = \frac{-\ln 0.24}{5} (\doteq 0.28542)$	1	
ii	$1 = 20e^{-kt}$ $e^{-kt} = 0.05$ $-kt = \ln 0.05$ $t = \frac{\ln 0.05}{-k}$ $= 10.49575 \doteq 10.5 \text{ hours}$	1 1	

MATHEMATICS - QUESTION NO: 14 continued

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

c)

i $\sin x = \cos x$

$\tan x = 1$

$x = \frac{\pi}{4}, \pi + \frac{\pi}{4}$

$\therefore A \left(\frac{\pi}{4}, \frac{1}{\sqrt{2}} \right)$ and $B \left(\frac{5\pi}{4}, -\frac{1}{\sqrt{2}} \right)$

ii $A = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x) dx$

$= \left[-\cos x - \sin x \right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$

$= \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right)$

$= \frac{4}{\sqrt{2}}$

$= 2\sqrt{2} \text{ u}^2$

1 (for finding the related angle $\frac{\pi}{4}$)

1

1

1

d)

i $A = 20000 (1.07)^{10}$

$= 39\ 343.02715$

$\approx \$39\ 343.03$

1

MATHEMATICS - QUESTION NO: 14 continued

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

d) ii Let A_1 be the value of the account on 01/07/2007

$$A_1 = 20000 \times 1.07 + 2000$$

$$A_2 = A_1 \times 1.07 + 2000$$

$$= 20000 \times 1.07^2 + 2000 \times 1.07 + 2000$$

$$A_3 = A_2 \times 1.07 + 2000$$

$$= 20000 \times 1.07^3 + 2000 \times 1.07^2 + 2000 \times 1.07 + 2000$$

⋮

$$A_{10} = 20000 \times 1.07^{10} + 2000 \times 1.07^9 + 2000 \times 1.07^8 + \dots$$

$$\dots 2000 \times 1.07 + 2000$$

$$= 20000 \times 1.07^{10} + 2000 (1 + 1.07 + 1.07^2 + \dots + 1.07^9)$$

GP with $a=1$, $r=1.07$ $n=10$

$$S_{10} = \frac{1(1.07^{10} - 1)}{1.07 - 1}$$

$$= 13.8164\dots$$

$$\therefore A_{10} = 39343.02715 + 27632.89592$$

$$= 66975.92307\dots$$

$$\div \$66975.92$$

1 for correctly identifying the general pattern

1 correct use of the formula for S_n of a G.P.

1 correct evaluation including both parts of the investment

MATHEMATICS - QUESTION NO: 14 continued

d)

iii $49565 = 20000(1+r)^{10}$

$$(1+r)^{10} = \frac{49565}{20000}$$

$$1+r = \sqrt[10]{\frac{49565}{20000}}$$

$$= 1.09500$$

$$\therefore r = 0.09500$$

$\therefore$ the interest rate is 9.5% p.a.

e)

let $y = \log_2 x^2$

$$= \frac{\ln x^2}{\ln 2}$$

$$= \frac{1}{\ln 2} \times \ln x^2$$

$$\therefore y' = \frac{1}{\ln 2} \times \frac{2x}{x^2}$$

$$= \frac{2}{x \ln 2}$$

$$\therefore \frac{d}{dx} \log_2 x^2 = \frac{2}{x \ln 2}$$

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

1

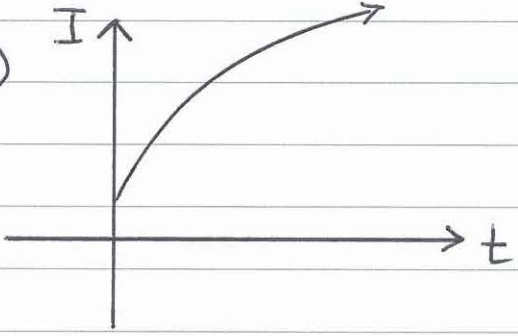
1

1

1

correct use of
change of base
- note that changing
to base 10 is of
no benefit, and
no marks are
awarded for this.

MATHEMATICS - QUESTION NO: 15 (18 marks)

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(a)(i) immigration was increasing $\therefore \frac{dI}{dt} > 0$	$\frac{1}{2}$	[CFPA means correct for previous answer]
(*tangent is positive) or $\frac{dI}{dt}$ is positive		
at a decreasing rate $\therefore \frac{d^2I}{dt^2} < 0$ (*curve is concave down) or $\frac{d^2I}{dt^2}$ is negative	$\frac{1}{2}$	
(ii) 	1	• $(-\frac{1}{2})$ if curve was below t-axis • If (i) wrong (no marks awarded), (ii) correct graph for (i) then ① mark was awarded.

MATHEMATICS - QUESTION NO: 15 continued

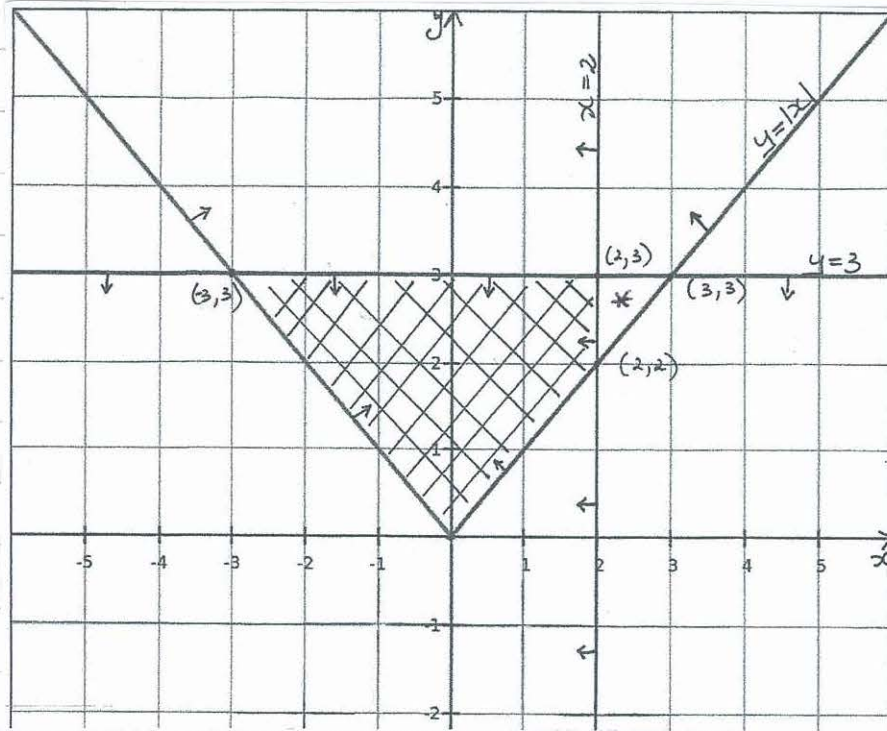
SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

(b)

(i)



*

2

correct region shaded

$1\frac{1}{2}$

lost $\frac{1}{2}$ mark if didn't have small triangle unshaded

$\frac{1}{2}$

region to the left of $x=2$

$\frac{1}{2}$

region below $y=3$

$\frac{1}{2}$

region inside $y=|x|$

$\frac{1}{2}$

correct intersection of all 3.

(ii) Area of Intersection

OR $A = \Delta + \text{trapezium}$

$= \text{Big } \Delta - \text{small } \Delta$

$= \frac{1}{2} \times 3 \times 3 + \frac{1}{2} \times 2(3+1)$

$= \frac{1}{2} \times 6 \times 3 - \frac{1}{2} \times 1 \times 1$

$= \frac{9}{2} + 4$

$= 9 - \frac{1}{2}$

$= 8\frac{1}{2}$ square units

$= 8\frac{1}{2}$ square units

$= \frac{17}{2}$ square units

$= \frac{17}{2}$ square units

OR equivalent.

1

correct answer

$\frac{1}{2}$

CFPA in (i)

as area was made easier to find

MATHEMATICS - QUESTION NO: 15 continued

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

(c)

• changing minutes to hours

cc) $\div 60$

t (min)	0	$\frac{1}{60}$	$\frac{2}{60}$	$\frac{3}{60}$	$\frac{4}{60}$	$\frac{5}{60}$ (hr)
V km/h	0	24	35	28	50	
$\div 60$	0	$\frac{2}{5}$	$\frac{7}{12}$	$\frac{7}{15}$	$\frac{5}{6}$	

• changing speed to km/min

$$\begin{aligned}
 x &= \int_0^{\frac{1}{15}} v \cdot dt \doteq \frac{1}{30} - 0 \left[0 + 4(24) + 35 \right] \\
 &\quad + \frac{1}{15} - \frac{1}{30} \left[35 + 4(28) + 50 \right] \\
 &= \frac{1}{180} \left[0 + 4(25) + 35 + 35 + 4(28) + 50 \right] \\
 &= \frac{1}{180} \times 328 \\
 &= \frac{328}{180} \text{ km} \\
 &= \frac{82}{45} \text{ km} \\
 &= 1.82 \text{ km}
 \end{aligned}$$

OR

$$\begin{aligned}
 x &= \int_0^{\frac{1}{15}} v \cdot dt \doteq \frac{1}{3} \left[0 + \frac{5}{6} + 4 \left(\frac{2}{5} + \frac{7}{15} \right) + 2 \left(\frac{7}{12} \right) \right] \\
 &= \frac{1}{3} \times \frac{82}{15} \\
 &= \frac{82}{45} \\
 &= 1.82 \text{ km}
 \end{aligned}$$

• one application
as per reference
sheet needs
3 function values

$$h = \frac{b-a}{n}$$

$$\therefore h = \frac{4-0}{4} = 1$$

(n subintervals
OR strips)

0 marks for wrong application of formula, ie $\frac{1}{3}$ between
0 & 24, there were no "inbetween" values given.

Majority of students
didn't read the
question carefully
to notice that
time and speed
were not both
in the same unit
of time

3 correct working
and answer

1 for changing minutes
to hours OR

changing speed to km/min

1 for substituting into
either formula
correctly

1 for correct answer

0 for wrong formula

MATHEMATICS - QUESTION NO: 15 continued

SUGGESTED SOLUTIONS		MARKS	MARKER'S COMMENTS																							
(c)	(c).		READ QUESTION!!! LEARN FORMULA!!!																							
	OR <table><tr><td>t</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>min</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>V</td><td>0</td><td>24</td><td>35</td><td>28</td><td>50</td></tr><tr><td>km/h</td><td>y_0</td><td>y_1</td><td>y_2</td><td>y_3</td><td>y_4</td></tr></table> FOTE * <u>not changing</u> <u>t</u> & <u>V</u>: $x = \int_0^4 v \cdot dt$ $\div \frac{1}{3} [0 + 4(24 + 28) + 2(35) + 50]$ $= \frac{1}{3} [4(52) + 2(35) + 50]$ $= \frac{1}{3} [208 + 70 + 50]$ $= \frac{328}{3} \text{ km} = 109.\dot{3} \text{ km} \quad \left\{ \begin{array}{l} 2 \text{ marks} \\ \text{out of 3} \end{array} \right\}$ ~~~~~ OR $x = \int_0^4 v \cdot dt \div \frac{2-0}{6} [0 + 4(24) + 35]$ $+ \frac{4-2}{6} [35 + 4(28) + 50]$ $= \frac{1}{3} [0 + 4(24 + 28) + 2(35) + 50]$ $= \frac{328}{3}$ $= 109.\dot{3} \text{ km} \quad \left\{ \begin{array}{l} 2 \text{ marks} \\ \text{out of 3} \end{array} \right\}$	t	0	1	2	3	4	min						V	0	24	35	28	50	km/h	y_0	y_1	y_2	y_3	y_4	2
t	0	1	2	3	4																					
min																										
V	0	24	35	28	50																					
km/h	y_0	y_1	y_2	y_3	y_4																					
		1	for correct working inside [] but wrong "fraction" outside.																							

MATHEMATICS - QUESTION NO: 15 continued

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(d) $16y = x^2 - 4x - 12$		• Learn axis of
(i) $y = \frac{x^2}{16} - \frac{4x}{16} - \frac{12}{16}$		Symmetry formula
$y = \frac{x^2}{16} - \frac{x}{4} - \frac{3}{4}$		$x = \frac{-b}{2a}$
$a = \frac{1}{16}$		• Learn how to
$b = -\frac{1}{4}$		complete the
$x = \frac{-b}{2a}$		square
$= - \frac{-\frac{1}{4}}{2(\frac{1}{16})}$		• Learn the
$\therefore x = 2$		importance of
$y = \frac{2^2}{16} - \frac{2}{4} - \frac{3}{4}$		the focal length.
$= -1 \therefore \text{Vertex } (2, -1)$	1	for finding
		axis of symmetry
		correctly $x=2$
	1	for $y=-1$
<u>OR</u> $x^2 - 4x + \left(-\frac{4}{2}\right)^2 = 16y + 12 + \left(-\frac{4}{2}\right)^2$		
$(x-2)^2 = 16y + 12 + 4$		
$= 16y + 16$	1	for completing the
$(x-2)^2 = 16(y+1)$		square to get to
$\therefore \text{Vertex } (2, -1)$	1	correct form.
		for correct
		vertex (or CFPA)

MATHEMATICS - QUESTION NO: 15 continued

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

(ii) needed $(x-2)^2 = 16(y+1)$ form

to find focal length

$$\text{ie } 4a = 16$$

$$a = 4$$

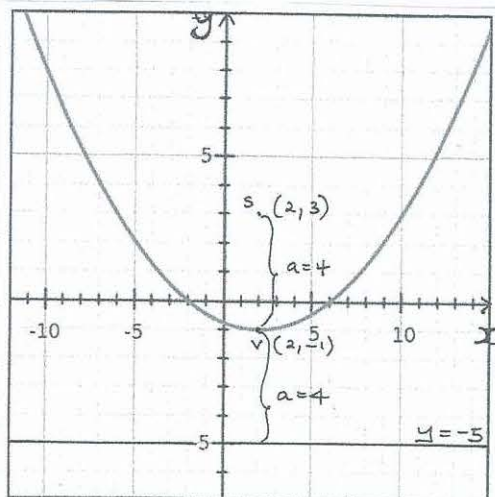
$\therefore$ Focus $(2, -1+4)$

$(2, 3)$

(iii) Directrix : $y = -1-4$

$$\therefore y = -5$$

A sketch will help to visualise the parabola and see where the focus & directrix are with respect to the focal length and vertex.



• Students that did a quick sketch didn't make errors.

$\frac{1}{2}$

if stated "a", then I was able to check focus & directrix (CFPA).

1

correct answer

1

correct answer

• Note, the directrix is a line not a point (no marks awarded if answer was given as a point)

MATHEMATICS - QUESTION NO: 15 continued

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(e) Learn basic formulas ! For cylinder : $V = \pi r^2 h$ $SA = 2\pi r^2 + 2\pi r h$		
(i) $V = 200$ $\pi r^2 h = 200$ $h = \frac{200}{\pi r^2}$	1	correct answer
(ii) $S = 2\pi r^2 + 2\pi r h$ $= 2\pi r^2 + 2\pi r \times \frac{200}{\pi r^2}$ $= 2\pi r^2 + \frac{400}{r}$	1	correct answer with working ie showing substitution of $h = \frac{200}{\pi r^2}$ and cancelling.

MATHEMATICS - QUESTION NO: 15 continued

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(iii) $S = 2\pi r^2 + 400r^{-1}$		
$\frac{dS}{dr} = 4\pi r - 400r^{-2}$	1	finds the first derivative of S
$= 4\pi r - \frac{400}{r^2}$		w.r.t. r .
Stationary values for S occur when $\frac{dS}{dr} = 0$		ie $\frac{dS}{dr}$ or S'
ie $4\pi r - \frac{400}{r^2} = 0$ ($r > 0$)		
$4\pi r = \frac{400}{r^2}$		
$4\pi r^3 = 400$		
$r^3 = \frac{400}{4\pi}$		
$r^3 = \frac{100}{\pi}$		
$r = \sqrt[3]{\frac{100}{\pi}}$	1	finds the zero of the first derivative,
$= 3.169202884$		ie $r = \sqrt[3]{\frac{100}{\pi}}$

MATHEMATICS - QUESTION NO: 15 continued

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

(e) (iii) continued

$$\frac{d^2s}{dr^2} = 4\pi + 800r^{-3}$$

$$= 4\pi + \frac{800}{r^3}$$

when $r^3 = \frac{100}{\pi}$, $\frac{d^2s}{dr^2} = 4\pi + \frac{800}{\frac{100}{\pi}}$

$$= 4\pi + 8\pi$$

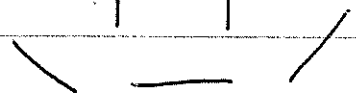
$$= 12\pi > 0$$

$$= 37.69911184 > 0 \quad \cup$$

$\therefore$ minimum area when $r = \sqrt[3]{\frac{100}{\pi}}$

OR

r	3	$\sqrt[3]{\frac{100}{\pi}}$	4
$\frac{ds}{dr}$	-6.74533	0	25.26548



$\therefore$ minimum area

will occur when

$$r = \sqrt[3]{\frac{100}{\pi}}$$

$$\therefore S = 2\pi \left(\sqrt[3]{\frac{100}{\pi}} \right)^2 + \frac{400}{\sqrt[3]{\frac{100}{\pi}}}$$

$$= 63.10735139 + 126.2147028$$

$$= 189.3220542$$

$$= 189 \text{ cm}^2 \text{ (to the nearest cm}^2\text{)}$$

1

Verifies this

Zero corresponds to a minimum S value, ie uses

$\frac{d^2s}{dr^2}$ or

equivalent, with numerical answers.

1

finds the value of S correctly to the nearest cm^2
ie $S = 189 \text{ cm}^2$